

Ibrahim Hussain

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Data Science Student | AI Systems, Data Engineering & Quantitative Modelling

Data Science student with experience in building AI systems, and data pipelines across finance, healthcare, and large-scale analysis. Skilled in time-series modelling, and multi-source data fusion, with a strong focus on extracting actionable insights from complex, high-dimensional datasets. Built and deployed 10+ projects including multi-agent AI systems, large-scale economic modelling, and multi-omics data integration, combining modern & traditional AI frameworks to solve real-world problems.

Experience

SMARTFORUM

AI Engineer Intern

June – September 2025

- Optimised RAG pipelines using PostgreSQL indexing and query optimisation, reducing retrieval latency by ~20% and improving response accuracy.
- Engineered modular multi-agent AI workflows using LangChain and LangGraph, automating core data-processing tasks and reducing manual intervention.
- Integrated Automatic Speech Recognition (ASR) enabling real-time interaction capabilities.
- Designed structured database schemas for efficient data storage, retrieval, and agent coordination in production workflows.

Projects & Technical Experience

EXPLAINABLE FINANCIAL RISK MANAGEMENT FRAMEWORK

AI Systems & Data Engineering Lead

January – April 2026

- Led end-to-end design and implementation of an explainable multimodal deep learning framework for financial risk management, combining market, SEC filings, macroeconomic, and graph-based data streams.
- Engineered large-scale financial data pipelines across 25 years of U.S. market data, including a final cleaned panel of 2,500 stocks $\times$ 6,286 trading days with zero missing values after multi-source filling and feature engineering.
- Processed SEC EDGAR fundamentals and textual filings, including 900k+ submissions JSON files, 17GB+ company facts, and 300k+ raw SEC filing documents.
- Designed the full model architecture and selection rationale, including a shared temporal attention encoder, FinBERT text encoder, sentiment/news analysts, StemGNN contagion module, regime risk module, drawdown module, liquidity risk module, and fusion layers.
- Adapted baseline graph neural network research models, including StemGNN and MTGNN, from price forecasting into specialised financial risk modules for contagion detection and market-regime identification.
- Built 3 leakage-safe chronological training splits across 2000–2024, using train/validation/test chunks to preserve temporal validity and improve model defensibility.
- Produced comprehensive technical documentation covering data engineering methodology, model-selection rationale, evaluation metrics, XAI design, and reproducible workflow decisions.

WRITER'S BLOCK

AI Engineer

December 2025

- Architected a resilient, multi-agent LLM system for automated, SEO-optimised content generation in the legal and healthcare sectors, reducing per-article cost to ~\$0.04.
- Engineered a production-grade content pipeline coordinating six parallel AI agents with auto-fallback models and retry logic, achieving an 85.5% success rate on generation tasks.
- Developed guardrails and a compiler agent to synthesise disparate LLM outputs into cohesive narratives and ensuring content quality.
- Built the full-stack application with Flask, and a Supabase/PostgreSQL database layer for real-time job tracking and data management.

TCGA INTEGRATIVE ANALYSIS

Data Engineer

November 2025

- Processed and integrated multi-omics cancer datasets (genomics, transcriptomics, epigenomics) with >20GB histopathology images.
- Engineered high-dimensional feature pipelines combining clinical, mutation, methylation, and gene expression data.
- Constructed unified modelling matrices combining mutation and epigenetic signals for downstream machine learning tasks.
- Built classification models to predict cancer stages and risk factors using structured biomedical datasets.

Education

FAST UNIVERSITY

Islamabad, Pakistan

Bachelor of Science in Data Science

September 2023 – Present

Relevant Coursework: Machine Learning, Deep Learning, Data Mining, Big Data Systems, Database Systems & Warehousing, Advanced Statistics, Operating Systems, Software Engineering.

Technical Skills

Programming Languages: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), SQL (PostgreSQL, MySQL), R, C++, x86 Assembly.

Data Science & Big Data Modelling: Time-Series Analysis, Regression (Ridge, Lasso, Elastic Net), Hypothesis Testing, Feature Engineering.

Machine Learning & AI: Deep Learning, Transformers, LangChain, LangGraph, BERT, Graph Neural Networks, Multi-Agent Systems.

Data Engineering: Large-Scale Data Pipelines, Time-Series Data Engineering, Data Cleaning, Missing Value Imputation, Multi-Source Data Fusion, Feature Engineering.

Quantitative Modelling: Financial Risk Modelling, Volatility, Drawdown Analysis, Portfolio Metrics, Regime Detection, Contagion Risk.

Interests: Economic Modelling, Graph Data Processing, Healthcare AI Research.

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